BIOLOGICAL SCIENCES 2B –GERHART
WINTER QUARTER 2026
MIDTERM #2 KEY

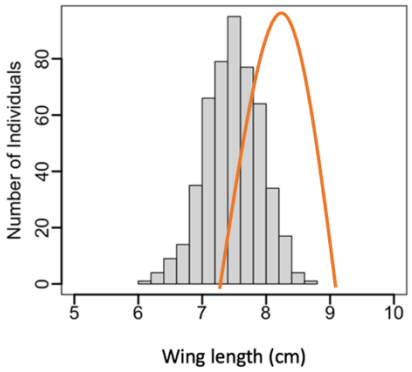
Note: On 'select all that apply' questions, partial credit is awarded in the following way: the total points for the question are split between all answer options. Students are awarded points for 1) correctly selecting a correct answer option, and 2) correctly ***not*** *selecting an incorrect answer option.*

1. All four answer options are possible fates of a mutation in a population
2. B: Because the number of gametes produced in the activity was small
3.
    a. C: Directional selection towards resistance
    b. B: Toxins in the environment
4. This individual can produce only *DF* and *dF* gametes
5. Gerharts with relative fitness less than 1 are Laci, Stephanie, Travis, Kelsey, Dillon, and Morgan
6. This cross will produce no heterozygote offspring. *Remember that genotype X^aY is not heterozygous, they are hemizygous*
7. A, C, and D are correct
8. f(short) = f(kk) = 0.473, so f(k) = sqrt(0.473) = 0.688 and f(K) = 1-0.688 = 0.312
   f(long) = f(KK) = 0.312^2 = **0.098**
9.
    a. Incomplete Dominance
    b. Pleiotropy
10. A, C, and D are correct
11. Cat #2 is homozygous recessive (because it has the recessive phenotype)
    Cat #9 is heterozygous (because it has the dominant phenotype and, when mating with homozygous recessive Cat #10, produces a homozygous recessive offspring, Cat #13)

12. In order to be the color they are and produce blue offspring, the magenta flowered plant must be *bbFF* and the white-flowered plant must be *BBff*, which then would produce only *BbFf* blue-flowered offspring.
    For full credit on this question, answers must correctly identify the sole correct genotype for each parent and the offspring; and clearly, completely, and accurately describe why these genotypes are the only ones possible for this cross.
13.
    a. You can conclude the following gene pairs are linked:
        - Hairy abdomen (a) and brown eyes (b)
        - Hairy abdomen (a) and dropped antennae (d)
        - Brown eyes (b) and dropped antennae (d)
          While data were not presented directly for the combination of hairy abdomen and brown eyes, we can conclude that these two genes are linked because hairy abdomen and dropped antennae are linked, and brown eyes and dropped antennae are linked; ergo, hairy abdomen and brown eyes must also be linked.
    b. Hairy abdomen (a) and dropped antennae (d) are closer together because their recombinant allele combinations (*Ad* and *aD*) occur in lower frequencies than for the other linked genes. *Since this question relied on identifying a pair of genes, partial credit was not awarded for correctly identifying one, but not the other, paired gene.*

14. Wing spots is dominant (and no wing spots is recessive). Parents 5 and 6 and their offspring 13 illustrate the dominance relationship. In order for both parents to have the same phenotype and produce offspring with a different phenotype, the parents must have the dominant phenotype (and be heterozygous) and their offspring has the recessive phenotype (having inherited the recessive allele from each heterozygous parent).
For full credit on this question, answers needed to clearly state which phenotype is dominant; identify the correct individuals from which the dominance can be determined; clearly, completely, and accurately explain why these individuals illustrate the dominance pattern; and include no other incorrect information. Remember that many 'tricks' for determining dominance from pedigrees (such as phenotypes skipping a generation, ratios of phenotypes in offspring, etc) are unreliable as they assume that 1) offspring numbers in humans are of sufficient sample size to follow Mendelian predictions (which is definitely not true) and/or 2) that the recessive allele is rare (which may or may not be true, and so cannot be assumed in all cases).

15. Since higher wing lengths have higher fitness (as shown in the right hand graph), we would expect phenotypes to shift to longer wings. For full points, curves needed to be clearly bell shaped (not linear, logistic, etc), show an appropriate rightward shift (centered over ~8.5 cm), and not suggest major declines in population size or increases in shorter wing lengths



16.

a. **Observed Genotypes:** f(BB) = 71/500 = 0.142 f(Bb) = 145/500 = 0.29
f(bb) = 284/500 = 0.568
Observed Alleles: f(B) = 0.142 + 0.5(0.29) = 0.287 f(b) = 0.568 + 0.5(0.29) = 0.713
Expected Genotypes: f(BB) = 0.287^2 = 0.0824 f(Bb) = 2(0.287)(0.713) = 0.4093
f(bb) = 0.713^2 = 0.5084

Allele	Observed Frequency
B	0.287
b	0.713

Genotype	Observed Frequency	Expected Frequency
BB	0.142	0.082
Bb	0.290	0.409
bb	0.568	0.508

b. This population is not in HWE. *For credit on this question, students needed to correctly interpret their table in part a, even if those values were incorrectly calculated. In order to interpret the table, at least one genotype needed to have both observed and expected values to compare.*

17.

a. 1/(2x43) = 1/86 = 0.0116 *Partial credit was awarded for calculating 1/43 = 0.0233*
b. 0.0000; the kitten is heterozygous and so does not display the recessive black phenotype

18. S = 61.23-56.27 = 4.96 R = 0.65 x 4.96 = 3.224 56.27 +3.224 = **59.494 cm horn length**